Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item

1. What is the main goal of the influence maximization problem in network diffusion? 1 / 1 point

- To determine which edges have the highest weight
- (selected) To select a set of initial nodes that maximizes the expected number of infected nodes
- To identify which nodes are most resistant to infection
- To prevent any node from becoming infected

Nice work

Feedback: Correct. The influence maximization problem seeks the seed nodes that maximize expected infection spread.

2. How does increasing modularity affect disease spread in the SIR model? 1 / 1 point

- It reduces both the degree and threshold of every node
- (selected) It often limits the epidemic to a single community
- It always speeds up the cascade across the whole network
- It ensures uniform diffusion across the network

Nice work

Feedback: Correct. High modularity restricts spread due to fewer inter-community edges.

3. What does the concept of submodularity capture in the context of the sigma function? 1 / 1 point

- That sigma always returns the same value regardless of seed nodes
- The relationship between edge weights and node degrees
- (selected) The idea that adding a new node to a smaller set yields more benefit than to a larger one
- The increase in infection rate with time

Nice work

Feedback: Correct. Submodularity reflects diminishing returns.

4. In the diffusion experiment using a physics collaboration network, which strategy achieved the highest cascade size? 1 / 1 point

- (selected) Greedy algorithm
- Degree centrality
- Random node selection
- Closeness centrality

Nice work

Feedback: Correct. The greedy strategy consistently yielded the largest cascades.

5. What conclusions can we draw from the vaccination strategy study? 1 / 1 point

- [x] High-degree nodes are more effective in low-modularity settings

Nice work

Feedback: Correct. Degree-based targeting performs better when connections are spread out.

- [ ] Both strategies perform equally well regardless of modularity
- [ ] Vaccination strategies are unaffected by network structure
- [x] High-betweenness nodes are the most effective to target in high-modularity networks

Nice work

Feedback: Correct. They act as bridges between communities.

6. In the influence maximization problem, what does the function sigma(A_0) represent? 1 / 1 point

- The actual number of nodes infected during a specific simulation
- The set of uninfected nodes in the network
- (selected) The expected number of infected nodes after the diffusion ends
- The probability that no nodes get infected

Nice work

Feedback: Correct. Sigma gives the expected value, accounting for stochasticity.

7. In the yeast molecular network study, what was the primary value of applying community detection? 1 / 1 point

- (selected) To reveal functional groupings such as protein complexes and dynamic units
- To track gene expression in real time
- To simulate random rewiring of the network
- To determine the evolutionary origin of each protein

Nice work

Feedback: Correct. Community detection identified meaningful biological groupings.

8. What does it mean for the sigma function to be monotonic? 1 / 1 point

- It decreases as more nodes are added
- It remains constant regardless of seed set changes
- (selected) It does not decrease when more nodes are added to the seed set
- It increases only when a seed node is removed

Nice work

Feedback: Correct. Monotonicity means that adding nodes cannot reduce the expected cascade size.

9. Which of the following are steps used by the greedy algorithm to build the seed node set? 1 / 1 point

- [ ] Randomly select new nodes at every step
- [x] Choose the node with the highest marginal gain in expected influence

Nice work

Feedback: Correct. Each node is selected based on its incremental contribution.

- [ ] Remove the least influential nodes in each iteration
- [x] Start with an empty set and add nodes one at a time

Nice work

Feedback: Correct. The greedy algorithm grows the seed set incrementally.

10. In high-modularity networks, which intervention strategy is most effective for epidemic control? 1 / 1 point

- Randomly selecting nodes for vaccination
- (selected) Vaccinating high-betweenness nodes to block cross-community paths
- Vaccinating nodes with the highest degree
- Targeting nodes with the lowest centrality

Nice work

Feedback: Correct. These nodes act as bridges, making them ideal intervention points.

11. What is a limitation of using the greedy algorithm for large networks? 1 / 1 point

- (selected) It is computationally expensive due to repeated simulations
- It always results in zero influence spread
- It cannot work on weighted graphs
- It requires exact knowledge of the optimal set

Nice work

Feedback: Correct. Simulations for every candidate make the process costly.

12. Why does modularity influence the effectiveness of vaccination strategies? 1 / 1 point

- It only affects degree-based measures, not betweenness
- It randomizes the location of the most connected nodes
- (selected) It determines whether inter-community connectors can be easily identified and blocked
- It increases the number of isolated nodes in the network

Nice work

Feedback: Correct. In highly modular networks, targeting bridges between communities is key.

13. Which of the following are properties that the sigma function must satisfy to apply the greedy algorithm? 1 / 1 point

- [x] Monotonicity

Nice work

Feedback: Correct. The function must be monotonic for the algorithm to work.

- [ ] Determinism
- [ ] Disconnectedness
- [x] Submodularity

Nice work

Feedback: Correct. Submodularity is key to the greedy approximation's performance guarantee.

14. Which of the following accurately describe the epidemic simulation results across varying modularity levels? 1 / 1 point

- [ ] Low-modularity networks never experienced cascades
- [ ] Cascade size always increases with modularity
- [x] Cascade speed peaked at a moderate modularity level in the linear threshold model

Nice work

Feedback: Correct. Speed was highest in a sweet spot of modularity.

- [x] Higher modularity led to smaller cascade sizes in the SIR model

Nice work

Feedback: Correct. Fewer inter-community edges limited the spread.

15. What is the main advantage of using centrality-based heuristics over the greedy algorithm? 1 / 1 point

- They are based on randomized node selection
- (selected) They are computationally cheaper to run
- They guarantee the optimal set of seed nodes
- They always result in higher cascade sizes

Nice work

Feedback: Correct. Centrality heuristics do not require costly simulations like the greedy approach.

16. Why is the nomination strategy considered effective in practice? 1 / 1 point

- [x] It captures central individuals through social nomination without needing the full network

Nice work

Feedback: Correct. Nominations tend to reach central nodes.

- [ ] It guarantees optimal diffusion in all cases
- [x] It performs well in real-world settings with limited network data

Nice work

Feedback: Correct. It's shown to be effective when full network mapping is impractical.

- [ ] It ensures completely uniform diffusion

17. Which of the following properties are essential for the greedy algorithm's performance guarantee? 1 / 1 point

- Directedness of the edges
- Fixed thresholds in all nodes
- (selected) Submodularity of the objective function
- Connectivity of the graph

Nice work

Feedback: Correct. Submodularity ensures diminishing returns, which underlies the guarantee.